

SHORING & EARTH RETENTION

PE Geotech

Soil nails · tiebacks · braced excavations — one-page reference

KEY EQUATIONS

$$Q_{all} = \frac{\pi D L_b q_u}{FS}$$

Pullout capacity. D = DRILLED HOLE dia (not the bar). L_b = bonded length, in the stable zone only. q_u = grout-ground bond stress. $FS \approx 1.5$ – 2.0

$$\theta = 45^\circ + \phi'/2$$

Failure-plane inclination from horizontal. Everything between the wall and this plane is the active wedge and carries NO pullout resistance.

$$T_{allow} = \frac{A_s f_y}{FS}$$

Steel tendon check. Capacity = LESSER of pullout and tendon. Always check both.

$$P_a = 0.65 K_a \gamma H$$

Peck apparent earth pressure, braced cuts in SAND (rectangular envelope). Clays use trapezoidal envelopes — not a simple Rankine triangle.

THE TRAPS (where the points actually go)

DIAMETER

Use the drilled-hole diameter, not the steel bar. Bond develops at the grout-to-soil boundary. Picking the bar can be $\sim 6\times$ low.

LENGTH

Use only the bonded length past the failure plane. The portion inside the wedge is anchored to the soil that is failing. Using full length runs high.

UNITS

Convert diameter to feet before it touches the equation. 6 in = 0.5 ft.

BOTH MODES

Grout can pull out OR steel can snap. The smaller governs.

WORKED EXAMPLE

25 ft wall · horizontal nail at 10 ft depth · total length 28 ft

$D = 6$ in hole (#8 bar) · $q_u = 3.0$ ksf · $\phi' = 32^\circ$ · $FS = 2.0$

1. $\theta = 45 + 32/2 = 61^\circ$
2. free length = $(25 - 10)/\tan 61^\circ = 8.3$ ft
3. $L_b = 28 - 8.3 = 19.7$ ft
4. $Q_{ult} = \pi(0.5)(19.7)(3.0) = 92.8$ kips
5. $Q_{all} = 92.8 / 2.0 = 46.4$ kips

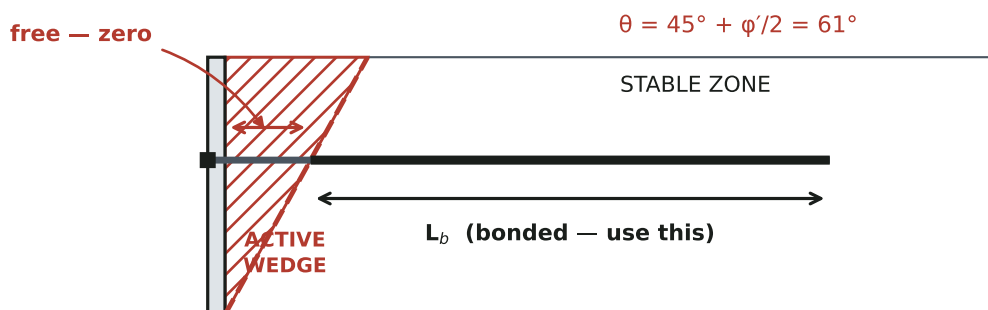
If you had used...

the BAR dia (1 in) → 7.7 kips (6× low)

the FULL 28 ft → 66.0 kips (42% high)

correct → **46.4 kips**

GEOMETRY



TYPICAL q_u (grout-ground)

loose sand	10–20 psi
stiff clay	25–45 psi
dense gravel	50–70 psi
caliche (LV)	100–150 psi
weathered rock	100–200 psi